

Computer Architecture- Quiz #4-G

Name: _____ Roll#: _____

Section: _____

Total Time: 20 minutes

Total Marks: 20

8th April, 2026

CLO 2:

Question 1

Marks 10

Consider the following **RISC-V** code. For each instruction, translate it into machine language and represent the result in **binary, hexadecimal** formats. You are also required to clearly identify and label all fields of machine instruction such as **opcode, rd, rs1, rs2, funct3, funct7, and immediate** required for machine tra

Address	Instructions																					
0x00000004	L1: or s0,s1,s2																					
	<table><tr><td></td><td>Funct7</td><td>RS2</td><td>RS1</td><td>Funct3</td><td>Rd1</td><td>opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary							Hex						
		Funct7	RS2	RS1	Funct3	Rd1	opcode															
	Binary																					
	Hex																					
0x00000008	addi t0,t1,-12																					
	<table><tr><td></td><td colspan="6">opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		opcode						Binary							Hex						
		opcode																				
	Binary																					
	Hex																					
0x0000000C	sw s3,-6(t2)																					
	<table><tr><td></td><td colspan="6">opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		opcode						Binary							Hex						
		opcode																				
	Binary																					
	Hex																					
0x00000010	lw t3,16(s0)																					
	<table><tr><td></td><td colspan="6">opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		opcode						Binary							Hex						
		opcode																				
	Binary																					
	Hex																					
0x00000014	bne t5,s0,L1																					
	<table><tr><td></td><td colspan="6">opcode</td></tr><tr><td>Binary</td><td colspan="6"></td></tr><tr><td>Hex</td><td colspan="6"></td></tr></table>		opcode						Binary							Hex						
		opcode																				
	Binary																					
	Hex																					

CLO 2:**Question 2****Marks 10**

Consider the following machine code and convert it to equivalent **RISC-V** code

Address	Instruction									
0x00000004	0x407302B3 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										
0x00000008	0x00C48913 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										
0x0000000C	0x00892383 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										
0x00000010	0xFE792D23 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										
0x00000014	0xFFE408E3 <table border="1"><tr><td></td><td></td><td>opcode</td></tr><tr><td>Binary</td><td colspan="2"></td></tr><tr><td>Decimal</td><td colspan="2"></td></tr></table> Final Instruction:			opcode	Binary			Decimal		
		opcode								
Binary										
Decimal										